

A Spectroscopic Analysis of Little Red Dots (LRDs)

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This project tested the active galactic nucleus (AGN) interpretation of Little Red Dots (LRDs) using James Webb Space Telescope data from the JWST Advanced Deep Extragalactic Survey (JADES) Data Release 3. LRD candidates were selected using redshift, color, and compactness criteria from the NIRCам photometric catalog, resulting in 24 candidates. Of these, 22 had available JWST/NIRSpec prism and/or grating spectra and were included in the spectroscopic analysis. Emission-line fitting was performed with PyQSOFit, focusing on lines such as $H\beta$, $[O\ III]\ \lambda 5007$, $[Ne\ III]\ \lambda 3869$, and $[Ne\ V]\ \lambda 3426$. Because $[Ne\ V]$ is a high-ionization line associated with AGN activity, the expected $[Ne\ V]$ flux was estimated from the measured $[O\ III]$ flux and compared to the local root-mean-square (RMS) noise near the $[Ne\ V]$ wavelength. None of the 22 sources reached a 3σ $[Ne\ V]$ detection threshold, so all $[Ne\ V]$ measurements were treated as non-detections and reported as upper limits. In 15 of the 22 sources, the expected $[Ne\ V]$ flux fell within the local 1σ noise level, making these cases especially inconclusive. The remaining 7 sources showed slightly more signal, with expected $[Ne\ V]$ fluxes above the local 1σ level, but still below the 3σ threshold. These results do not confirm $[Ne\ V]$ emission in the sample, but they place upper-limit constraints. These upper limits are crucial for clarifying what the spectra can and cannot rule out about obscured AGN activity. Currently, I am working on reducing data for nine additional sources from Alessandro Peca's June 2026 observations and setting up a reduction pipeline with PyPeIt. More objects will be observed in the coming months.